LESSON PLAN

AIR NAVIGATION (21)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	18
Total No. of Lesson Plans	18	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Training on use of advanced test equipment: - Digital & analog multimeter & Megger, Digital & analog Insulation tester, Bonding tester and No volt tester
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Training on use of advanced test equipment: - Digital & analog multimeter & Megger, Digital & analog Insulation tester, Bonding tester and No volt tester.
5. Specific Objectives	21.1.11.1 Trainee will be able to explain / practically use advanced test equipment in aircraft maintenance.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical demonstration and hands-on operation of Digital/Analog Multimeter, Megger, Insulation Tester, Bonding Tester, and No Volt Tester with safety precautions and instructor supervision.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a multimeter?
(b) Why is insulation resistance testing critical in aircraft systems?
(c) What safety precaution is essential before using a Megger?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Digital and Analog Multimeter Operation • Megger and Insulation Tester Principles • Bonding Tester and No Volt Tester Usage	L & Ds	PPT, IB	15 min
(b)	• Safety Precautions for High Voltage Testing • Calibration and Zeroing Procedures • Interpretation of Test Readings and Faults	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting Common Test Equipment Errors • Maintenance and Storage of Test Instruments • Documentation of Test Results	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a Megger and a standard multimeter.
(b) Describe the correct procedure for testing aircraft bonding continuity.
(c) What steps must be taken after completing an insulation resistance test?

12. Home Assignment and Follow Up	Prepare a brief report on the specific safety protocols required when operating a Megger on live aircraft systems, referencing the next lesson plan (LP 02) on electrical system troubleshooting.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min